

Seeing with ‘Fresh Eyes’: Development and Validation of the Fresh Experiences Scale

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BACKGROUND

- Evidence suggest that the acute subjective effects experienced under psychedelics are the strongest predictors of long-term therapeutic outcomes¹
- There remains under-explored phenomena within psychedelic experiences²
- Currently no psychometric tool measure the phenomena of seeing the familiar in new perspectives

Cellular Changes
Psychoplasticity Model,
Critical Period Re-opening

Neural Circuit Changes
Predictive Processing Theory,
REBUS Model,
Reduced DMN Connectivity

Psychological Changes
Psychological Flexibility,
Cognitive Flexibility,
Openness to Experience

Phenomena of
Fresh Experiences

Qualitative reports of *fresh experiences* in clinical studies³:

“Being among the trees were incredible, like experiencing them for the first time, so vibrant, so alive”

“I noticed things I didn’t notice usually, the leaves on the trees and the birds, small details”

“I would look over at the park, and it would be so green, a type of green I’d never experienced before”

RESEARCH AIM

To identify a conceptually and statistically parsimonious set of items for the Fresh Experiences Scale (FES), and to demonstrate the scale's reliability and validity through survey-based psychometric evaluation

METHODOLOGY

Item Development

- Expert Group Study (*n* = 5) to assess face validity
- Feedback used to revise and refine items
- Preliminary FES developed contained 51-items

Retrospective Survey Study

- Convenience sampling using online survey
- n* = 410 participants answered based on a selected past psychedelic experience

Exploratory Factor Analysis

- Exploratory sample (*n* = 260) to identify latent constructs
- Poorly correlated items removed prior to EFA

Confirmatory Factor Analysis

- Validation sample (*n* = 150) to test model fit of structure obtained in EFA
- Assessed using several fit indices

Validity Analysis

- Total sample (*n* = 410) to evaluate discriminant, convergent and predictive validity of final FES
- Correlation and regression analyses against relevant psychometric measures

Reliability Analysis

- Assess internal consistency of final FES items
- Test-Retest Reliability assessed using ICC in a follow-up sample (*n* = 120)

RESULTS

DEMOGRAPHICS

- Mean Age: 32 years (SD = 11.4)
- Education Level: 68.8% with an undergraduate degree or higher
- Gender: 61.7% males, 33.7% females
- Psychedelic used: LSD (39.5%) and Psilocybin (43.4%) most common

1 EXPLORATORY FACTOR ANALYSIS

- 25 items removed due to low correlation
- Scree plot supported 2 factor structure
- 11 items removed in EFA for parsimony

Item Description	F1	F2	h ²
I noticed the particular characteristics of some aspects of myself.	0.78		0.49
Some familiar experiences, things or people acquired a new meaning.	0.77		0.61
I noticed many details about my thoughts or feelings.	0.71		0.42
I saw familiar things or people through fresh eyes.	0.63		0.54
I noticed the distinctive characteristics of a given thing or person.	0.62		0.48
I thought about things in a new way.	0.57	0.26	0.61
It felt like I was seeing certain things or people for the first time.	0.44	0.22	0.38
I felt enchanted, intrigued, or captivated.		0.89	0.62
My experience felt rich.		0.85	0.66
My experience felt fulfilling.		0.76	0.57
My experience felt full or complete.		0.70	0.49
I found my experiences interesting.		0.66	0.51
There was a child-like freshness to my experiences.		0.63	0.60
The world took on a feeling of newness.	0.25	0.57	0.60
I felt a child-like sense of wonder.		0.57	0.48
Cumulative Variance Explained	0.26	0.54	
Eigen values	8.25	1.29	

Figure 1: EFA result of final set of FES items

2 CONFIRMATORY FACTOR ANALYSIS

Factor 1
Fresh Framing
 $\alpha = 0.84$
 $\omega = 0.90$

Factor 2
Fresh Feeling
 $\alpha = 0.88$
 $\omega = 0.92$

Cognitive and perceptual reframing of the familiar in new ways

Feelings of heightened salience or richness

	RMSEA [90% CI]	CFI	TLI	SRMR	χ^2 /df
MLR Estimator					
Two Factor Model	.101 [.079-.122]	.875	.853	.072	1.99*
One Factor Model	.116 [.095-.136]	.834	.806	.077	3.15*
WLSMV Estimator					
Two Factor Model	.051 [.026-.071]	.952	.943	.070	1.38
One Factor Model	.062 [.042-.081]	.927	.915	.077	1.20

Figure 2: Goodness-of-fit metrics of final FES items using two different estimators (MLR and WLSMV)

3 VALIDITY ANALYSES

FES measures aspects **distinct** from:

- Intensity of experience (mDEQ)
- Increased energy or pep (Vitality Subscale of SF-36)

FES aligns closely with:

- Gaining of new insights (PIQ)
- Changed meaning of perceptions (11d-ASC)

	FES	FFeel	Discriminant Intensity (mDEQ)	Vitality Subscale (SF-36)	Convergent Insightfulness (11d-ASC)	Changed Meaning of Percepts (11d-ASC)	PIQ Total
FES Total	-						
FFeel	0.94***	-	0.29***	0.27***	0.63***	0.57***	0.56***
FFrame	0.91***	0.71***	0.33***	0.13**	0.61***	0.58***	0.62***

FES associated with positive changes in wellbeing attributed to psychedelic experience

PTCS Items	Standardised Estimate	Standard Error	p-value	r ²
Positive Mood/Emotional Experiences	0.410	0.040	< 0.001***	0.239
Negative Mood/Emotional Experiences	-0.326	0.049	< 0.001***	0.106
Self Esteem/Self-worth	0.406	0.046	< 0.001***	0.178
Positive Social Effects	0.422	0.044	< 0.001***	0.207
Problematic/Addictive Behaviours	-0.238	0.052	< 0.001***	0.059
Health-related Behaviours	0.211	0.046	< 0.001***	0.053

4 RELIABILITY ANALYSES

FES demonstrated good internal consistency and test-retest reliability

	Intraclass Correlation	95% Confidence Interval	F Test	p-value
		Lower Bound	Upper Bound	
Single Rater	0.81	0.74	0.87	9.8
Average Rater	0.90	0.85	0.93	9.8

Test-Retest Reliability Bland-Altman plot: FES

Figure 3: Differences in mean FES scores between main survey (T1) and follow-up survey (T2)

DISCUSSION

- First validated scale targeting the phenomenology of new perspectives under psychedelics
- Two-factor structure aligns well with mechanistic accounts of how psychedelics confer therapeutic benefit
- FES demonstrated good reliability and construct validity; model requires further validation
- Fresh experiences may be predictive of positive well-being outcomes

Scan here for the Final FES

FUTURE DIRECTIONS

- Cross-validate findings in studies with larger sample sizes
- Explore relationships between fresh experiences and mechanistic accounts
- Evaluate utility of FES in clinical studies for predicting therapeutic outcomes

References:

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